

Checkpoint

14.4: Tangent Planes and Normal Lines

Problem 1. *What is a level curve to a surface $f(x, y)$?*

Problem 2. *The gradient of a k -dimensional surface lives in what dimension? What does the gradient vector represent with respect to a level curve of the surface?*

Problem 3. *If a plane needs a point and a normal? What will we use for the normal of a **tangent** plane?*

Problem 4. *If $2(x - 1) + 3(y - 2) + 4(z - 3) = 0$ is a tangent plane to some surface $f(x, y)$, find the corresponding normal line at the same point.*

Problem 5. *If $\nabla f(x, y, z) = \langle xy, x + y + z, x - y - z \rangle$ find the equation of the tangent plane at $(1, 2, 3)$.*

1 Answers

1. It is setting our surface equal to a constant $f(x, y) = c$.
2. $k - 1$. The gradient is the normal vector to the level curve.
3. We use the gradient of a surface one dimension higher by thinking about our surface as a specific level surface of something larger.
4. $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$.
5. $2(x - 1) + 6(y - 2) - 4(z - 3) = 0$